Test Series No.

1.4

MASTERY TEST SERIES

DURATION: 90 Minutes | **M. MARKS: 400**

DATE: 13/02/2026

SYLLABUS

Physics:	Electric Charges and Fields, Electrostatic Potential and Capacitance, Current Electricity, Moving Charges and Magnetism + 30% Questions from cumulative syllabus of previously conducted part test
Chemistry:	Solutions, Electrochemistry, Chemical Kinetics, Haloalkanes and Haloarenes + 30% Questions from cumulative syllabus of previously conducted part test
Botany:	Plant Growth and Development, Sexual Reproduction in Flowering Plants + 30% Questions from cumulative syllabus of previously conducted part test
Zoology:	Chemical Coordination and Integration, Animal kingdom, Human Reproduction + 30% Questions from cumulative syllabus of previously conducted part test"

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **90 minutes** duration and the Test Booklet contains **100** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 400.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

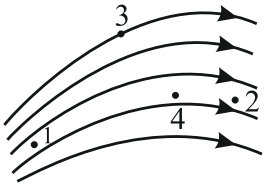
Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Observe the given electric field line diagram. At which point is the electric field strength the highest?



(1) Point 1 (2) Point 2
(3) Point 3 (4) Point 4

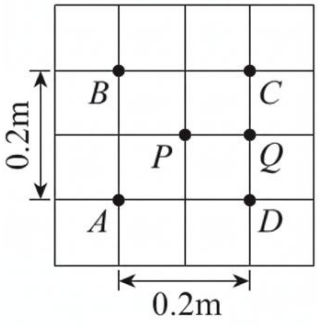
2. According to Newton, the viscous force acting between liquid layers of area A depends on velocity gradient = $\frac{\Delta v}{\Delta z}$ (Δv = change in velocity, Δz = thickness of liquid layer). Force can be expressed as $F = -\eta A\frac{\Delta v}{\Delta z}$ where η is a constant. The dimension of η is

(1) $[ML^2T^{-2}]$
(2) $[ML^2T^{-1}]$
(3) $[ML^{-1}T^{-1}]$
(4) $[M^0L^0T^0]$

3. Two balls are thrown vertically upward, one after the other, at an interval of 2 seconds. Both balls are thrown with an initial velocity of 40 m/s along the same vertical line. At what height above the ground will the two balls collide? (Take g = 10 m/s²)

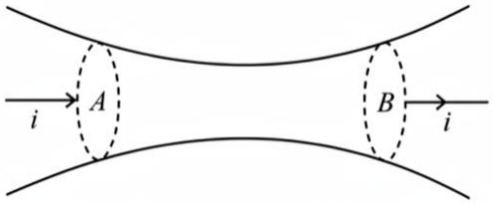
(1) 60 m
(2) 75 m
(3) 80 m
(4) 120 m

4. *A, B, C, D, P* and *Q* are points in a uniform electric field. The potentials at these points are $V(A) = 2$ volt, $V(P) = V(B) = V(D) = 5$ volt, $V(C) = 8$ volt. The electric field at *P* is:



(1) 10 Vm^{-1} along PQ
(2) 15 $\sqrt{2}$ V m^{-1} along PA
(3) 5 V m^{-1} along PC
(4) 5 V m^{-1} along PA

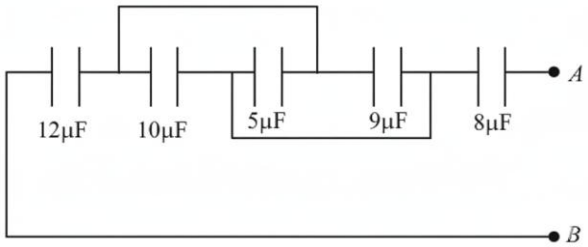
5. A constant current *i* is flowing along the length of a non-uniform cross-section of conducting wire as shown in figure.



The current density along wire in moving from A to B is:

(1) first increases then decreases.
(2) first decreases then increases.
(3) remains constant.
(4) initially increases then becomes constant.

6. Five capacitors together with their capacitances are shown in the adjoining figure. The potential difference between the points A and B is 60 volt. The equivalent capacitance between the points A and B and charge on 5μF capacitor will be respectively–



Assume the connecting wires have negligible resistance and the circuit is in a steady state.

(1) 44 μF, 30 μC (2) 16 μF, 150 μC
(3) 15 μF, 200 μC (4) 4 μF, 50 μC

7. A charge *q* is initially (at $t = 0$ s) moving with a velocity $\vec{v} = (3\hat{i})$ m/s. If magnetic field and electric field in the region is $(5\hat{j})$ Tesla and $(-15\hat{k})$ N/C respectively, then velocity (in m/s) of charge *q* at $t = 3$ s is:

(1) zero (2) $-5\hat{j}$
(3) $15\hat{k}$ (4) $3\hat{i}$

8. A uniform resistance wire of 0.5 ohm per cm is bent to form a square loop PQRS with each side measuring 8 cm. A wire of the same material and thickness connects the diagonal corners Q and S. A 4 V battery with negligible internal resistance is connected across the vertices P and R. The total current drawn from the battery is:

(1) 0.5 A (2) 1.0 A
(3) 2.0 A (4) 4.0 A

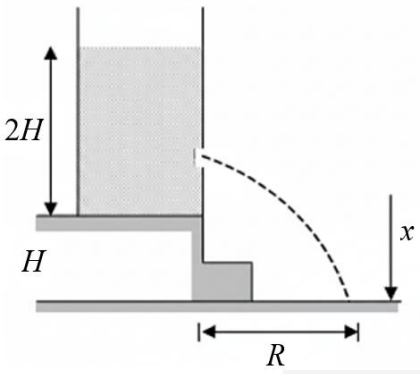
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9. Two thin long parallel wires separated by a distance b are carrying a current i ampere each. The magnitude of the force per unit length exerted by one wire on the other is:

(1) $\frac{\mu_o i^2}{b^2}$ (2) $\frac{\mu_o i^2}{2\pi b}$

(3) $\frac{\mu_o i}{2\pi b}$ (4) $\frac{\mu_o i}{2\pi b^2}$

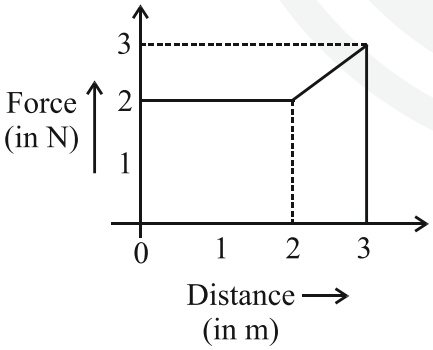
10. A tank is filled upto a height $2H$ with a liquid and is placed on a platform of height H from the ground. The distance x from the ground where a small hole is punched to get the maximum range R is:



(1) H (2) 1.25 H

(3) 1.5 H (4) 2 H

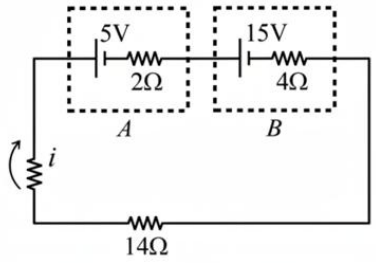
11. A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3 m is



(1) 4 J (2) 2.5 J

(3) 6.5 J (4) 5 J

12. Current i is flowing in a circuit as shown in figure.



Internal resistance of cell A and cell B are 2 Ω and 4 Ω respectively. Match **List-I** (current and potential difference) with **List-II** (numeric value).

List-I		List-II	
(A)	Current i in the circuit (in ampere)	(I)	14
(B)	Terminal potential difference across cell A (in volt)	(II)	3
(C)	Potential difference across 14 Ω resistance (in volt)	(III)	11
(D)	Terminal potential difference across cell B (in volt)	(IV)	1

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-I, D-III

(2) A-III, B-IV, C-II, D-I

(3) A-II, B-I, C-IV, D-III

(4) A-II, B-IV, C-I, D-III

13. If the distance between the earth and the sun were half its present value, the number of days in a year would have been:

(1) 64.5 (2) 129

(3) 182.5 (4) 730

14. Two rods of different materials having coefficients of thermal expansion α_1, α_2 and Young's moduli Y_1, Y_2 respectively are fixed between two rigid massive walls. The rods are heated such that they undergo the same increase in temperature. There is no bending of the rods. If $\alpha_1 : \alpha_2 = 2:3$, the thermal stresses developed in the two rods are equal provided $Y_1 : Y_2$ is equal to:

(1) 2 : 3 (2) 1 : 1

(3) 3 : 2 (4) 4 : 9

15. Two simple harmonic motions are represented by:

$$y_1 = 3\left[\sin 2\pi t + \sqrt{3}\cos 2\pi t\right]$$

and $y_2 = 3\sin\left(2\pi t + \frac{\pi}{4}\right)$

The ratio of the amplitudes of y_1 and y_2 is

(1) 1 : 1 (2) 2 : 1

(3) 1 : 3 (4) $\sqrt{3}:1$

16. Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C till the temperature of the calorimeter and its contents rises to 80°C. The mass of the steam condensed in kg is:

(1) 0.130 (2) 0.065

(3) 0.260 (4) 0.165

17. When a current of 5 mA is passed through a galvanometer having a coil of resistance 15 Ω, it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into a voltmeter of range 0–10 V is:

(1) 2.045×10^3 Ω (2) 2.535×10^3 Ω

(3) 4.005×10^3 Ω (4) 1.985×10^3 Ω

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18. A charge q is placed at the centre of the line joining two equal charges Q. The system of the three charges will be in equilibrium if q is equal to:

(1) $-\frac{Q}{2}$ (2) $-\frac{Q}{4}$

(3) $+\frac{Q}{4}$ (4) $+\frac{Q}{2}$

19. Two identical capacitors, have the same capacitance C. One of them is charged to potential V_1 and the other to V_2. Likely charged plates are then connected. Then, the decrease in energy of the combined system is:

(1) $\frac{1}{4}C(V_1^2 - V_2^2)$

(2) $\frac{1}{4}C(V_1^2 + V_2^2)$

(3) $\frac{1}{4}C(V_1 - V_2)^2$

(4) $\frac{1}{4}C(V_1 + V_2)^2$

20. Two fuses are rated at 10 A and 5 A. The two fuse wires have equal but small resistance. If they are joined in parallel, what will be the effective rating of the combination?

(1) 15 A (2) 10 A

(3) 7.5 A (4) 5 A

21. A charged particle moves in a circle perpendicular to uniform magnetic field. The time period of revolution of particle is independent of

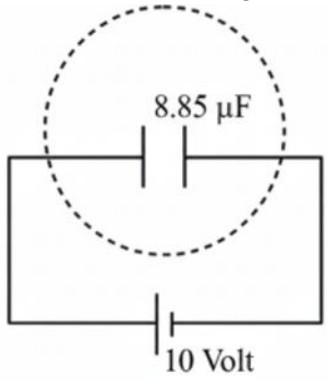
(1) Magnetic field

(2) Charge of particle

(3) Mass of the particle

(4) Speed of the particle

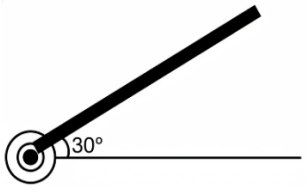
22. A charged parallel plate capacitor is enclosed inside a spherical closed surface as shown in figure. The net electric flux through the spherical surface is:



(1) 10^7 V-m (2) 10^6 V-m

(3) 10^3 V-m (4) Zero

23. A rod of length 50 cm is pivoted at one end. It is raised such that it makes an angle of 30° from the horizontal as shown and released from rest. Its angular speed when it passes through the horizontal (in rad s^{-1}) will be: (Take, g = 10 ms^{-2})



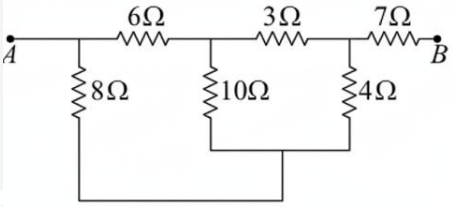
(1) $\frac{\sqrt{30}}{2}$

(2) $\sqrt{30}$

(3) $\frac{\sqrt{20}}{3}$

(4) $\frac{30}{2}$

24. The equivalent resistance between the points A and B is:



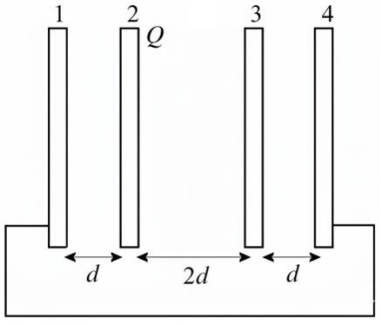
(1) $\frac{36}{7}\Omega$

(2) 10 Ω

(3) $\frac{85}{7}\Omega$

(4) 5 Ω

25. Four metallic plates are placed as shown in the figure. Plate 2 is given a charge Q whereas all other plates are uncharged. Plates 1 and 4 are joined together. The area of each plate is same.



The charge appearing on the right side of plate 3 is

(1) zero

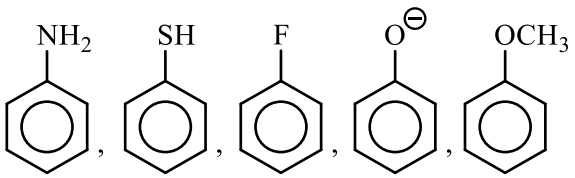
(2) +Q/4

(3) –3Q/4

(4) Q/2

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26. How many groups (attached with benzene ring) can show +M effect?

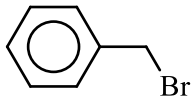


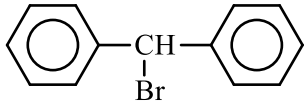
(1) 2
(2) 3
(3) 4
(4) 5

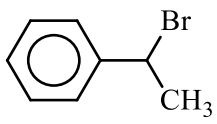
27. The van't Hoff factor (i) for 0.1 M $MgCl_2$ solution is 2.8. The degree of dissociation of $MgCl_2$ is:
(1) 85%
(2) 80%
(3) 95%
(4) 90%

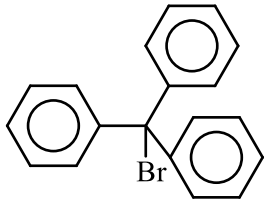
28. The slope of a graph between ln k and $\frac{1}{T}$ for Arrhenius equation is:
(1) $\frac{2.303R}{E_a}$
(2) $\frac{-E_a}{2.303R}$
(3) $\frac{R}{E_a}$
(4) $\frac{-E_a}{R}$

29. Among the following most reactive compound towards S_N2 reaction is:

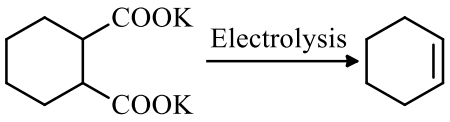
(1) 

(2) 

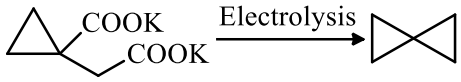
(3) 

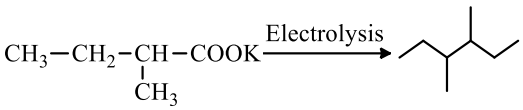
(4) 

30. In which of the following reaction, the Kolbe's electrolysis product is **incorrectly** mentioned?

(1) 

(2) 

(3) 

(4) 

31. Given below are two statements:
Statement I: The negative terminal in an electrochemical cell is cathode.
Statement II: The potential of standard hydrogen electrode may be positive, negative or zero.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

32. Match **List-I** with **List-II**.

List-I (Name reaction)		**List-II (Product)**	
(A)	Wurtz Fittig reaction	(I)	Diphenyl
(B)	Fittig reaction	(II)	Chlorobenzene
(C)	Finkelstein reaction	(III)	Methylbenzene
(D)	Sandmeyer's reaction	(IV)	Ethyl iodide

Choose the **correct** answer from the options given below:
(1) A-III; B-I; C-IV; D-II
(2) A-IV; B-I; C-III; D-II
(3) A-I; B-IV; C-II; D-III
(4) A-IV; B-III; C-II; D-I

33. The freezing point of 1 molal aq. NaCl solution is:
(Given: K_f of H_2O is 1.86 K kg mol^{-1})
(1) –3.72°C (2) –1.86°C
(3) 0°C (4) –2.79°C

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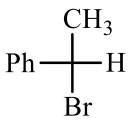
34. In paper chromatography, the stationary phase is:
(1) Ethanol
(2) Silica gel
(3) Water trapped in cellulose
(4) Mobile solvent

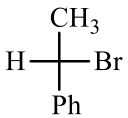
35. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Λ_m of CH_3COOH changes with dilution.
Reason R: On dilution degree of dissociation of CH_3COOH increases.
In the light of above statements, choose the **correct answer** from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

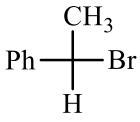
36. Number of chlorine atoms in one molecule of DDT is:
(1) 3 (2) 4
(3) 5 (4) 6

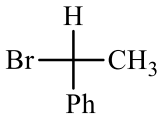
37. How long will 100 g of reactant will take to reduce to 25 g if the rate constant is $2 \times 10^{-2}\ s^{-1}$?
[Given: log 2 = 0.301]
(1) 69.3 s
(2) 13.8 s
(3) 84.9 s
(4) 33.7 s

38. The pair of enantiomers among the following compound is:

I. 

II. 

III. 

IV. 

(1) I and IV
(2) II and IV
(3) II and III
(4) I and II

39. The vapour pressure of pure liquids A and B are 450 and 700 mm Hg respectively, at 350 K. Find out the composition of the liquid mixture if total vapour pressure is 600 mm Hg.
(1) $\chi_A = 0.6$ and $\chi_B = 0.4$
(2) $\chi_A = 0.3$ and $\chi_B = 0.7$
(3) $\chi_A = 0.4$ and $\chi_B = 0.6$
(4) $\chi_A = 0.7$ and $\chi_B = 0.3$

40. E_{cell} of the given cell is:
$Mg(s) + 2Ag^+(10^{-2}M) \rightarrow Mg^{2+}(10^{-3}M) + 2Ag(s)$,
$E^o_{cell} = 3.17\ V$
(1) 3.11 V
(2) 3.19 V
(3) 3.14 V
(4) 3.21 V

41. How many different alkenes with molar mass 84 upon hydrogenation can give 3-methyl pentane?
(1) 3
(2) 4
(3) 5
(4) 6

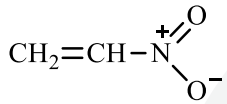
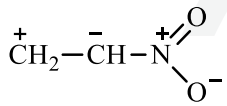
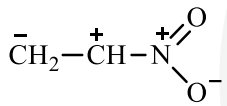
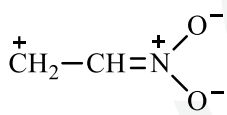
42. The experimental data for the reaction, P + Q → R is as follows:

Expt. No.	[P] ($mol\ L^{-1}$)	[Q] ($mol\ L^{-1}$)	Rate ($mol\ L^{-1}\ s^{-1}$)
1.	0.5	0.5	1.5×10^{-5}
2.	0.5	1.0	3.0×10^{-5}
3.	1.0	1.0	3.0×10^{-5}

The overall order of reaction is:
(1) 3
(2) 1
(3) 2
(4) 0

[Contd...

43. Given below are two statements:
Statement I: S_N2 reactions are stereospecific.
Statement II: S_N1 reactions involve formation of carbocations.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

44. Least contributing resonating structure of nitro ethane is:
(1) 
(2) 
(3) 
(4) 

45. The molality of 49% H_2SO_4 (mass/volume) solution if the density of the solution is 1.49 g/mL, is:
(1) 0.2 m
(2) 2 m
(3) 0.5 m
(4) 5 m

46. Which of the following solutions has minimum boiling point?
(1) 0.1 m Na_2SO_4
(2) 0.01 m $(NH_4)_3PO_4$
(3) 0.5 m $BaCl_2$
(4) 0.2 m $Al_2(SO_4)_3$

47. Given below are two statements:
Statement I: Molecularity is applicable only for elementary reactions.
Statement II: For complex reaction molecularity has no meaning.
In the light of the above statements, choose the **most appropriate** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

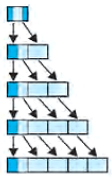
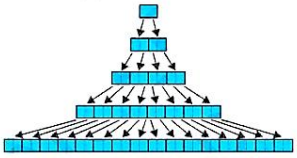
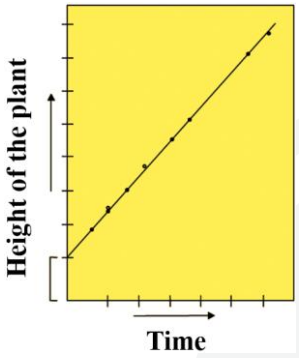
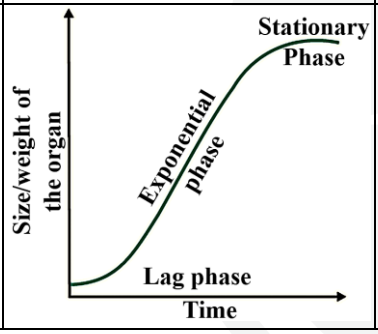
48. The vapour pressure of a solution (in mm Hg) containing 18 g of fructose ($C_6H_{12}O_6$) in 180 g of water is:
[Given: p^0 = Vapour pressure of pure water]
(1) $\frac{p^0}{101}$
(2) $\frac{p^0}{100}$
(3) $\frac{101\,p^0}{100}$
(4) $\frac{100\,p^0}{101}$

49. The molar conductance of $\frac{M}{32}$ solution of a weak monobasic acid is 8.0 mho cm^2 mol^{-1} and at infinite dilution is 400 mho cm^2 mol^{-1}. The dissociation constant of this acid is:
(1) 1.25×10^{-5}
(2) 1.25×10^{-6}
(3) 1.25×10^{3}
(4) 1.25×10^{-4}

50. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The addition of HX to alkenes is a nucleophilic addition reaction.
Reason R: In the first step, the alkene attacks the proton of HX.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

[Contd...

51. Match **List-I** with **List-II** w.r.t to growth.

List-I		List-II	
(A)		(I)	Constant linear growth
(B)		(II)	An idealised sigmoid growth curve
(C)		(III)	Arithmetic growth
(D)		(IV)	Geometric Growth

Choose the *most appropriate* answer from the options given below:

(1) A-III, B-IV, C-I, D-II

(2) A-II, B-I, C-III, D-IV

(3) A-II, B-I, C-IV, D-III

(4) A-III, B-IV, C-II, D-I

52. When the ovules are borne on the central axis and septa are absent, the placentation is called:

(1) axile as in *Dianthus*.

(2) parietal as in *Argemone*.

(3) free central as in *Primrose*.

(4) marginal as in bean.

53. Which of the following equations **correctly** represents exponential growth?

(1) $L_t = L_0 + rt$ (2) $W_1 = W_0 e^{rt}$

(3) $W_1 = W_0 + rt$ (4) $L_t = L_0 \times rt$

54. Given below are two statements.

Statement-I: In basidiomycetes, sexual spores are generally not found, but vegetative reproduction by fragmentation is common.

Statement-II: In ascomycetes asexual spores are called ascospores which are produced endogenously in sac like asci.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement-I and Statement-II are correct

(2) Both Statement-I and Statement-II are incorrect

(3) Statement-I is correct and Statement-II is incorrect

(4) Statement-I is incorrect and Statement-II is correct

55. In gymnosperms, the highly reduced male gametophyte, which is confined to only a limited number of cells, is called a:

(1) microsporophyll.

(2) pollen grain.

(3) microsporangium.

(4) male strobilus.

56. In angiosperms when a pollen grain is shed at the 3-celled stage, it contains:

(1) two vegetative cells and one generative cell.

(2) one vegetative cell and one generative cell.

(3) one vegetative cell and two male gametes.

(4) two generative cells and one male gamete.

57. Read the following statements (A-E) regarding auxin.

A. Auxins induce parthenocarpy.

B. Auxin controls xylem differentiation.

C. Auxin helps in cell division.

D. 2, 4-D is widely used to kill monocotyledonous weeds.

E. Auxins like IAA and indole butyric acid (IBA) have been isolated from plants.

How many of the above statements are **correct**?

(1) Four

(2) One only

(3) Two only

(4) Three only

58. In dicot stems:

(1) hypodermis is sclerenchymatous.

(2) the innermost layer of the cortex is rich in starch grains.

(3) water cavities are present.

(4) phloem parenchyma is absent.

59. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Measurement and the comparison of total growth per unit time is called the absolute growth rate.

Reason (R): The growth of the given system per unit time expressed on a common basis, e.g., per unit initial parameter is called the relative growth rate.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

60. The two polar nuclei in the angiosperm embryo sac are located:

(1) inside the egg cell.

(2) at the chalazal end within the antipodal cells.

(3) in the large central cell.

(4) between the two synergids.

61. Identify the **mismatched** pair regarding the electron transport system (ETS):

(1) Complex II — Succinate dehydrogenase

(2) Cytochrome c — Mobile carrier attached to the inner surface of the inner membrane

(3) Complex III — Cytochrome bc_1 complex

(4) Complex IV — Contains cytochromes a, a_3 and two copper centres

62. Match **List-I** with **List-II**:

List-I (Plant Growth Regulator)		**List-II (Specific Function)**	
(A)	Ethylene	(I)	Hastens maturity in juvenile conifers
(B)	Abscisic Acid	(II)	Overcomes apical dominance
(C)	Cytokinin	(III)	Initiates germination in peanut seeds
(D)	Gibberellin	(IV)	Inhibits seed germination

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-I, C-II, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-IV, C-I, D-III

63. Which of the following plants do not produce both chasmogamous and cleistogamous flowers?

(1) Common pansy (2) *Oxalis*

(3) *Commelina* (4) Papaya

64. Taxonomic category Class includes:

(1) various phyla.

(2) related orders.

(3) related kingdoms.

(4) related division.

65. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Mitosis usually results in the production of diploid daughter cells with non-identical genetic complement.

Reason (R): Mitotic divisions in the meristematic tissues – the apical and the lateral cambium, result in a continuous growth of plants throughout their life.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

[Contd...

66. Apomixis is:
(1) a form of sexual reproduction that mimics asexual reproduction.
(2) a form of asexual reproduction that mimics sexual reproduction.
(3) observed in banana.
(4) fruit development without fertilisation.

67. Identify the **correct** statements regarding plasma membrane.
A. Depending on the ease of extraction, membrane proteins can be classified as integral and peripheral.
B. The ratio of protein and lipid varies considerably in different cell types.
C. In human beings, the membrane of the erythrocyte has approximately 40 per cent protein and 52 per cent lipids.
D. The major lipids are phospholipids that are arranged in a bilayer.
E. The lipids are arranged within the membrane with the polar head towards the outer sides and the hydrophobic tails towards the inner part.
Choose the **correct** options:
(1) A, B, C and E only
(2) A, B, C and D only
(3) B, C, D and E only
(4) A, B, D and E only

68. Identify the **correct** statement.
(1) Embryos of monocotyledons possess only one cotyledon.
(2) In the grass family the endosperm is called scutellum.
(3) The portion of the embryonal axis below the level of attachment of scutellum is the epicotyl.
(4) Epicotyl has a root apex and a few leaf primordia.

69. How many turns of the Calvin cycle are required for the formation of three molecules of glucose?
(1) Three (2) Six
(3) Twelve (4) Eighteen

70. Which plant is commonly known as "carrot grass" and has become ubiquitous in occurrence?
(1) *Vallisneria* (2) *Hydrilla*
(3) *Parthenium* (4) *Zostera*

71. Given below are two statements.
Statement-I: Turgidity of cells helps in extension growth.
Statement-II: Water provides the medium for enzymatic activities needed for growth.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement-I and Statement-II are correct
(2) Both Statement-I and Statement-II are incorrect
(3) Statement-I is correct and Statement-II is incorrect
(4) Statement-I is incorrect and Statement-II is correct

72. How many of the following show water pollination?
(A) Water lily
(B) *Zostera*
(C) *Vallisneria*
(D) *Hydrilla*
Choose the **correct** option:
(1) One only
(2) Two only
(3) Three only
(4) Four

73. Identify the **correct** statement regarding kinetin.
(1) It is a natural cytokinin found in corn kernels.
(2) It was isolated from human urine.
(3) It does not occur naturally in plants.
(4) It was isolated from coconut milk

74. The hard outer layer of the pollen grain called the exine is made up of:
(1) cellulose. (2) sporopollenin.
(3) galactans. (4) pectin.

75. Growth in a dorsiventral leaf is best measured in terms of:
(1) increase in length.
(2) increase in volume.
(3) increase in surface area.
(4) increase in cell number.

[Contd...

76. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Immunity is decreased in old individuals.
Reason R: Thymus gland is degenerated in old individuals which leads decreased production of thymosins and weakened immune response.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

77. Choose the **correct** statements regarding Adrenal hormones:
A. Adrenaline and Noradrenaline are called catecholamines.
B. They decrease the heart beat and rate of respiration.
C. They are secreted in response to stress.
D. They stimulate the breakdown of glycogen into glucose.
(1) A, B and C (2) A, C and D
(3) B, C and D (4) A, B and D

78. Given below are two statements.
Statement I: The 8th, 9th and 10th pairs of ribs are called floating ribs.
Statement II: Ribs are bicephalic because they have two articulation surfaces on the dorsal end.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

79. Identify the **correct** sequence of the male accessory ducts starting from the seminiferous tubules.
(1) Rete testis → Vasa efferentia → Epididymis → Vas deferens
(2) Vasa efferentia → Rete testis → Epididymis → Vas deferens
(3) Rete testis → Vas deferens → Epididymis → Vasa efferentia
(4) Vasa efferentia → Epididymis → Rete testis → Vas deferens

80. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Hypothalamus	(I)	Controls respiration and gastric secretions
(B)	Cerebellum	(II)	Urge for eating and drinking
(C)	Corpus callosum	(III)	Convoluted surface for more neurons
(D)	Medulla	(IV)	Connects cerebral hemispheres

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-II, B-I, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-III, C-IV, D-II

81. Which class of Vertebrata is characterized by pneumatic bones?
(1) Reptilia
(2) Amphibia
(3) Aves
(4) Mammalia

82. Which of the following are autoimmune disorders?
A. Myasthenia gravis
B. Muscular dystrophy
C. Osteoporosis
D. Gout
(1) A only
(2) A and B only
(3) A and C only
(4) A, B, C and D

83. Radial symmetry is found in which of the following groups?
(1) Annelids and Arthropods
(2) Sponges and Platyhelminthes
(3) Coelenterates, Ctenophores and Echinoderms
(4) Molluscs and Chordates

84. Identify the joint between the atlas and axis.
(1) Ball and socket joint
(2) Hinge joint
(3) Pivot joint
(4) Gliding joint

[Contd...

85. Given below are two statements.
Statement I: The pars intermedia is almost merged with pars distalis in humans.
Statement II: Pars intermedia secretes several trophic hormones including MSH.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

86. Consider the following statements regarding the hormones during pregnancy:
A. hCG, hPL, and relaxin are produced in women only during pregnancy.
B. Relaxin is only secreted by the placenta in the later phase of pregnancy.
C. During pregnancy, levels of other hormones like estrogens and progestogens are increased several folds in maternal blood.
D. Increased levels of these hormones are essential for supporting foetal growth.
Which of the above statements are **correct**?
(1) A, B, and C only
(2) A, C, and D only
(3) B and D only
(4) A, B, C and D

87. In which subphylum of Chordata does the notochord extend from head to tail region and persist throughout their life?
(1) Urochordata (2) Cephalochordata
(3) Vertebrata (4) Both (1) and (2)

88. Which of the following statements about the human female reproductive system are **true**?
A. The oviducts, uterus, and vagina constitute the female accessory ducts.
B. The womb is shaped like an upright pear and is supported by ligaments.
C. The endometrium undergoes cyclical changes during the menstrual cycle.
D. The myometrium exhibits strong contraction during delivery of the baby.
(1) A, C, and D only
(2) B and C only
(3) A and D only
(4) A and B only

89. Members of class Cyclostomata are unique among vertebrates because they:
(1) have a sucking and circular mouth with jaws.
(2) lack scales and paired fins but migrate to fresh water for spawning.
(3) possess a bony vertebral column and scales.
(4) are endoparasites on larger marine mammals.

90. The electrical potential difference across the resting plasma membrane is called:
(1) resting potential
(2) nerve impulse
(3) action potential
(4) depolarization

91. Match **List-I** with **List-II**.

List-I (Animal)		**List-II (Feature)**	
(A)	*Sycon*	(I)	Comb plates
(B)	*Pleurobrachia*	(II)	Choanocytes
(C)	*Pila*	(III)	Radula
(D)	*Balanoglossus*	(IV)	Proboscis gland

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-III, D-IV

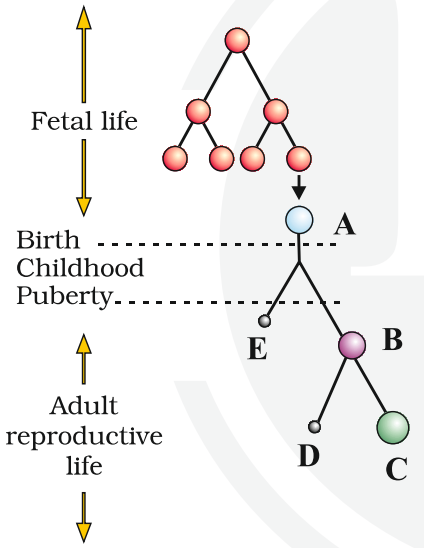
92. Which cells in the neural system can detect, receive, and transmit stimuli?
(1) Schwann cells
(2) Neurons
(3) Glial cells
(4) Nissl's granules

93. Identify the hypercalcemic hormone that increases blood Ca^{2+} levels.
(1) Thyrocalcitonin
(2) Thymosins
(3) Parathyroid hormone
(4) Glucagon

94. White muscle fibers differ from Red muscle fibers in having:
(1) High myoglobin content.
(2) Large number of mitochondria.
(3) Dependency on aerobic process for energy.
(4) High amount of sarcoplasmic reticulum.

95. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The endometrium is necessary for the implantation of the fertilized ovum and other events of pregnancy.
Reason R: In the absence of fertilization, the corpus luteum degenerates, leading to the disintegration of the endometrium.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

96. In the given diagrammatic representation of oogenesis, which of the following cells are haploid?



(1) C and D only
(2) B and C only
(3) B, C, D and E only
(4) A, B, C, D and E

97. Which hormone is responsible for the milk ejection from the mammary glands?
(1) Prolactin (2) Oxytocin
(3) Vasopressin (4) LH

98. Animals of which of the following phyla is triploblastic but acoelomate?
(1) Aschelminthes (2) Platyhelminthes
(3) Annelida (4) Porifera

99. Match **List-I** with **List-II**.

List-I (Cells/Structure)		**List-II (Function/Characteristic)**	
(A)	Leydig Cells	(I)	Provide nutrition to germ cells
(B)	Sertoli Cells	(II)	Formed after the first meiotic division
(C)	Primary Spermatocytes	(III)	Periodically undergo meiosis
(D)	Secondary Spermatocytes	(IV)	Synthesise and secrete androgens

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-III, D-II
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-I, C-II, D-III
(4) A-I, B-IV, C-III, D-II

100. A middle-aged patient presents with severe disfigurement, especially of the face. This condition is **most** likely:
(1) Gigantism caused by over-secretion of GH in childhood.
(2) Acromegaly caused by excess secretion of GH in adults.
(3) Dwarfism caused by low secretion of GH.
(4) Goitre caused by iodine deficiency.

■■■

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